FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

. The first line of input contains an integer representing the number of pages in the reference string.
The second line contains the page reference string as a sequence of integers.
The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

```c
#include <stdio.h>

int main() {
    int n, frames_count, i, j;
    int page_faults = 0, page_hits = 0, pointer = 0;

    printf("Enter number of pages in reference string: \n");
    if (scanf("%d", &n) != 1) return 0;

    int reference_string[n];
    printf("Enter the reference string: \n");
    for (i = 0; i < n; i++) {
        scanf("%d", &reference_string[i]);
    }

    printf("Enter number of frames: \n");
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Completed

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Select Language : C (GCC 9.2.0)

```c
    if (scanf("%d", &frames_count) != 1) return 0;

    int frames[frames_count];
    for (i = 0; i < frames_count; i++) {
        frames[i] = -1;
    }

    for (i = 0; i < n; i++) {
        int found = 0;

        for (j = 0; j < frames_count; j++) {
            if (frames[j] == reference_string[i]) {
                found = 1;
                page_hits++;
                break;
            }
        }

        if (!found) {
```

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```c
            if (frames[j] == reference_string[i]) {
                found = 1;
                page_hits++;
                break;
            }
        }

        if (!found) {
            frames[pointer] = reference_string[i];
            pointer = (pointer + 1) % frames_count;
            page_faults++;
        }
    }

    printf("\nTotal Page Faults: %d", page_faults);
    printf("\nTotal Page Hits: %d\n", page_hits);

    return 0;
}
```

Test Cases 2

Run Sample Cases

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> Custom Test

Sample Test Case

> Test Case - 1